



Checking and securing solving linear equations

Task A: Solving equations involving a bracketed expression

1) Spot the mistake in each method to solve an equation involving brackets.

a)

$$5(x + 10) = 55$$

$$5x + 10 = 55$$

$$5x = 45$$

$$x = 9$$

The brackets have not been expanded properly.

The second line should be $5x + 50 = 55$

b)

$$2(4x + 10) = 18$$

$$2x + 5 = 9$$

$$2x = 4$$

$$x = 2$$

When dividing the LHS by 2 the terms in the bracket should not be divided. The second line should be $4x + 10 = 9$

c)

$$7(x - 3) = 42$$

$$x - 3 = 6$$

$$x = 3$$

The additive inverse of -3 is $+3$ so 3 should have been added to both sides. Line 3 should be $x = 6 + 3$ or $x = 9$

d)

$$5(3x + 2) = 14$$

$$8x + 10 = 14$$

$$8x = 4$$

$$x = 2$$

The brackets have not been expanded properly 5 and 3 should be multiplied not added. The second line should be $15x + 10 = 55$

e)

$$\frac{1}{3}(2x - 5) = 15$$

$$2x - 5 = 3$$

$$2x = 8$$

$$x = 4$$

The reciprocal of a third is three. The first step should be to multiply by 3 not divide by 3.

f)

$$\frac{3}{4}(10 - 8x) = 6$$

$$\frac{30}{4} - \frac{24}{4}x = 6$$

$$-\frac{24}{4}x = -\frac{3}{2}$$

$$-x = -\frac{3}{2} \times \frac{4}{24} \quad x = -\frac{3}{2} \times \frac{4}{24}$$

$$x = \frac{1}{4} \quad x = -\frac{1}{4}$$

It would have been easier to multiply by the reciprocal first.



2a) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$6(x - 9) = 21$$

$$6x - 54 = 21$$

$$6x = 75$$

$$x = \frac{75}{6} = \frac{25}{2}$$

The second method requires a lot more fractions skills so the first method is likely to be more efficient.

Multiplying by the reciprocal first:

$$6(x - 9) = 21$$

$$x - 9 = \frac{21}{6} = \frac{7}{2}$$

$$x = \frac{7}{2} + 9$$

$$x = \frac{7}{2} + \frac{18}{2}$$

$$x = \frac{25}{2}$$

2b) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$15(4x + 7) = 45$$

$$60x + 105 = 45$$

$$60x = -60$$

$$x = -1$$

This is personal preference. Both expressions have a common factor of 15 so dividing by 15 first makes the values smaller and potentially easier to work with compared to expanding the brackets first.

Multiplying by the reciprocal first:

$$15(4x + 7) = 45$$

$$4x + 7 = 3$$

$$4x = -4$$

$$x = -1$$

2c) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$\frac{6}{5}(3x + 4) = 30$$

$$\frac{18}{5}x + \frac{24}{5} = 30$$

$$\frac{18}{5}x = 30 - \frac{24}{5}$$

$$\frac{18}{5}x = \frac{126}{5}$$

$$x = \frac{126}{5} \times \frac{5}{18}$$

$$x = 7$$

It is much easier to 'undo' the multiplicative step first this time as $30 \times \frac{5}{6}$ is an integer

Multiplying by the reciprocal first:

$$\frac{6}{5}(3x + 4) = 30$$

$$3x + 4 = 30 \times \frac{5}{6}$$

$$3x + 4 = 25$$

$$3x = 21$$

$$x = 7$$



Task B: Solving equations with multiple brackets

1) The solutions to the equations are given. Fill in the steps to show how to get to these answers. *Other methods may also be valid.*

a)

$$5(x + 6) = 7(9x - 4)$$

$$5x + 30 = 63x - 28$$

$$30 = 58x - 28$$

$$58 = 58x$$

$$x = 1$$

b)

$$2(4x - 5) = -6(3x - 7)$$

$$4x - 5 = -3(3x - 7)$$

$$4x - 5 = -9x + 21$$

$$13x = 26$$

$$x = 2$$

c)

$$8(x - 5) + 3(x + 5) = 8$$

$$8x - 40 + 3x + 15 = 8$$

$$11x - 25 = 8$$

$$11x = 33$$

$$x = 3$$

d)

$$4(3x - 1) - 2(3x - 1) = 22$$

$$2(3x - 1) = 22$$

$$3x - 1 = 11$$

$$3x = 12$$

$$x = 4$$

e)

$$2(4x - 9) - 4(3 - 2x) = 50$$

$$8x - 18 - 12 + 8x = 50$$

$$16x - 30 = 50$$

$$16x = 80$$

$$x = 5$$

f)

$$3(2(x + 4) + 5) = 75$$

$$3(2x + 8 + 5) = 75$$

$$3(2x + 13) = 75$$

$$2x + 13 = 25$$

$$2x = 12$$

$$x = 6$$

2) Solve these equations twice. Firstly by expanding first, secondly by simplifying before expanding. Write a sentence to compare the efficiency.

a) $4(x + 5) + 3(x + 5) = 14$ $4x + 20 + 3x + 15 = 14$ $7x + 35 = 14$ $7x = -21$ $x = -3$	<i>E.g. Both methods were reasonably efficient. Simplifying first meant that a common factor could be seen which eliminated a step.</i>	$4(x + 5) + 3(x + 5) = 14$ $7(x + 5) = 14$ $x + 5 = 2$ $x = -3$
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b) $21 + 3(2x - 3) = 7(2x - 3) - 3(2x - 3)$ $21 + 6x - 9 = 14x - 21 - 6x + 9$ $6x + 12 = 8x - 12$ $12 = 2x - 12$ $24 = 2x$ $x = 12$	<i>E.g. Simplifying first seemed to be more efficient. No expanding brackets was necessary and less manipulating negative numbers was needed.</i>	$21 + 3(2x - 3) = 7(2x - 3) - 3(2x - 3)$ $21 + 3(2x - 3) = 4(2x - 3)$ $21 = 2x - 3$ $24 = 2x$ $x = 12$
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